NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Monitoring and control



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CONCEPTS COVERED

- Monitoring earned value



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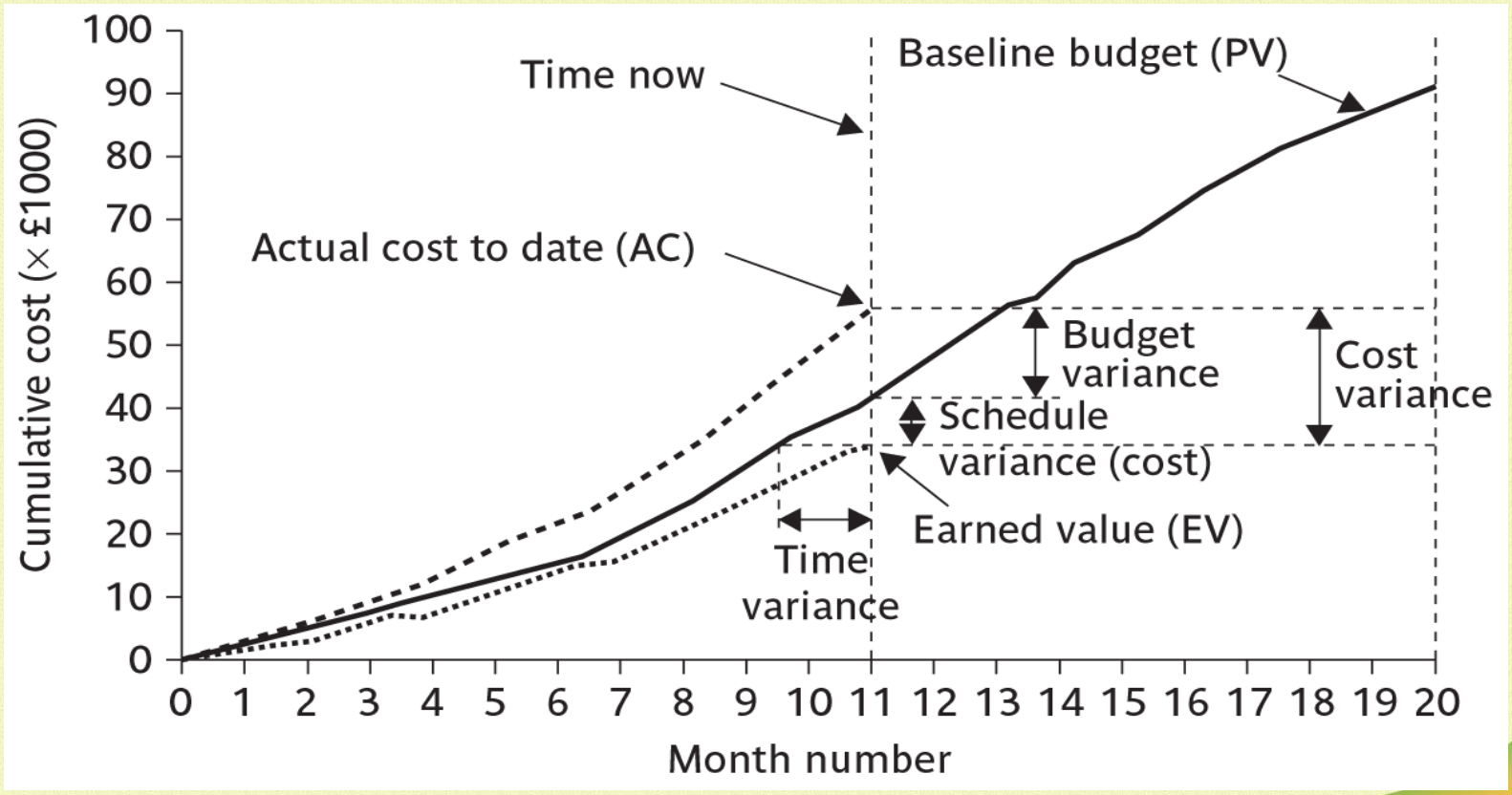
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Monitoring earned value

- Having created the baseline budget, the next task is to monitor earned value as the project progresses.
- This is done by monitoring the completion of tasks (or activity starts and milestone achievements in the case of the other crediting techniques).
- As well as recording EV, the actual cost of each task can be collected as *actual cost* (AC).
- This is also known as the *actual cost of work performed* (ACWP).





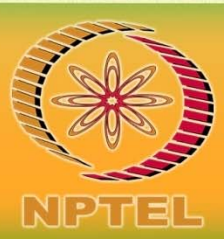


An earned value tracking chart

Monitoring earned value

- The figure illustrates the following performance statistics, which can be shown directly or derived from the earned value.
  - Schedule variance (SV)
  - Time variance (TV)
  - Cost variance (CV)
  - Performance ratios





Schedule variance (SV)

- The schedule variance is measured in cost terms as SV = EV-PV
- It indicates the degree to which the value of completed work differs from that planned.
- Say, for example, that work with a PV of £40,000 should have been completed by now. In fact, some of that work has not been done, so that the EV is only £35,000.
- So, SV = £35,000- £40,000 = - £5,000.
- A negative SV means the project is behind schedule.





Time variance (TV)

- This is the difference between the time when the achievement of the current earned value was planned to occur and the time now.
- In this case, the current EV should have been achieved in the early part of month 9 and as the time now is the end of month 11,
  - So, the TV is about -1.75 months.
  - TV negative indicates project is running late.

Cost variance (CV)

- This is calculated as CV = EV-AC
- It indicates the difference between the earned value or budgeted cost and the actual cost of completed work.
- Say that when the SV above was calculated as -£5,000, £55,000 had actually been spent to get the EV.
- The CV in this case would have been £35,000- £55,000, i.e. - £20,000.
- It can also be an indicator of the accuracy of the original cost estimates.
- A negative CV means that the project is over cost.

Performance ratios

- Two ratios are commonly tracked:
  - The cost performance index (CPI=EV/AC)
  - The schedule performance index (SPI=EV/PV)
- Using the examples,
  - CPI would be £35,000/ £55,000, that is, 0.64
  - SPI would be £35,000/ £40,000, that is, 0.88
- The two ratios can be thought of as a 'value-for-money' indices.
- A value greater than one indicates that work is being completed better than planned, whereas a value of less than one means that work is costing more than and/or proceeding more slowly than planned.





Performance ratios cont ...

- CPI can be used to produce a revised cost estimate for the project, called estimate at completion (EAC).
- EAC is calculated as BAC/CPI where BAC (budget at completion) is the current projected budget for the project.
- If the BAC was £100,000 then a revised estimate at completion (EAC) would be £100,000/0.64 or £156,250.
- Similarly, the current SPI can be used to project the possible duration of the project given the current rate of progress.





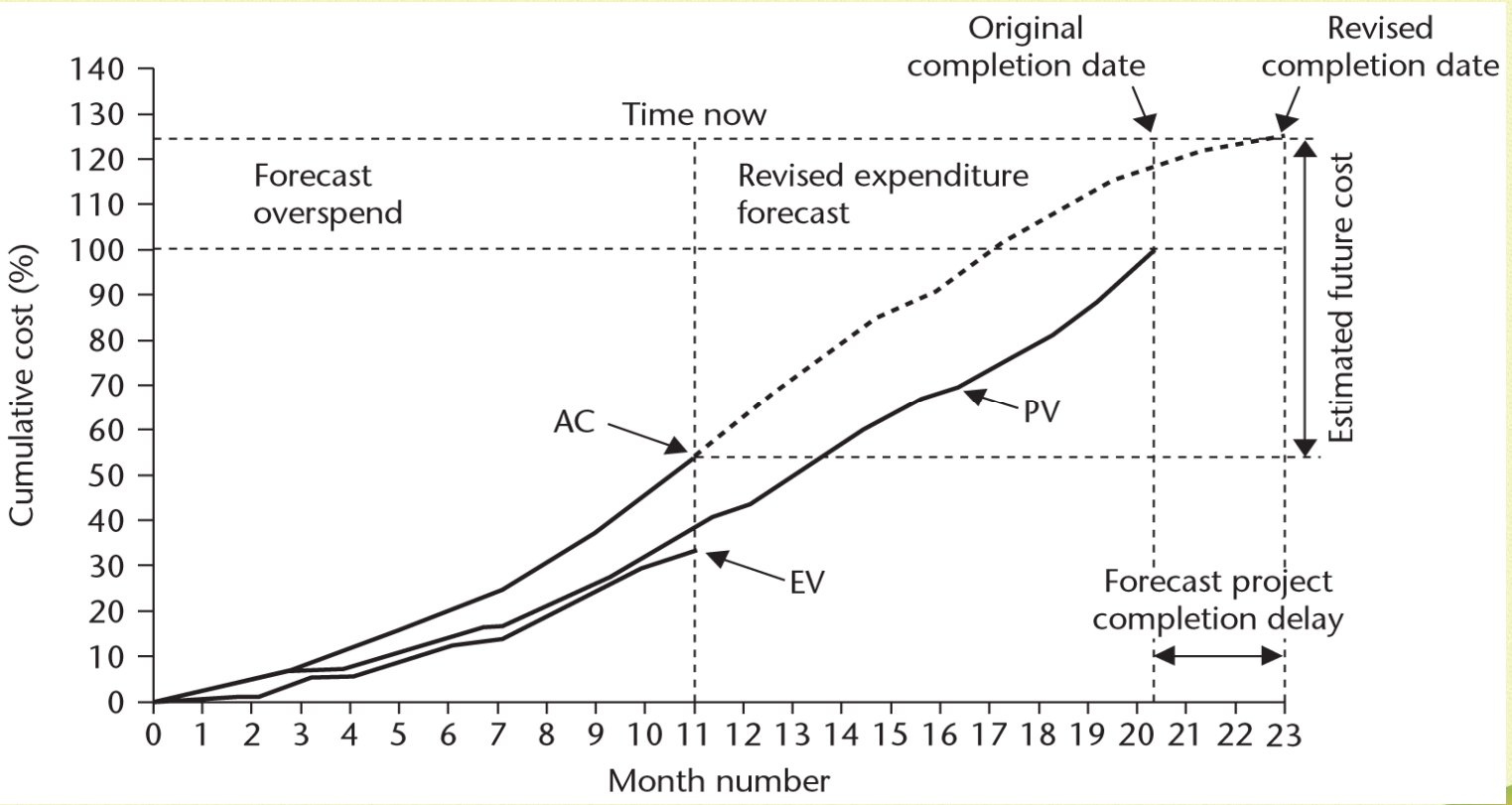
Performance ratios cont ...

- Say the planned total duration is 23 months – in earned value terminology this is called *schedule at completion* (SAC).
- A time estimate at completion (TEAC) can be calculated as SAC/SPI.
- In this case, it would be 23/0.88, that is, 26.14 months.
- This is only an approximate guide:
  - Where there are several parallel chains of activities being carried out concurrently
    - the project duration will depend on the degree to which the activities that have been delayed are on the critical path.





An earned value chart with revised forecasts



Performance ratios cont ...

- Earned value analysis has not yet gained universal acceptance for use with software development projects,
  - perhaps largely because of the attitude that, whereas a half-built house has value reflected by the labour and materials that have been used,
    - a half-completed software project has virtually no value at all.
- This is to misunderstand the purpose of earned value analysis, which is a method for tracking what has been achieved on a project
  - Measured in terms of the budgeted costs of completed tasks or products.





Earned value – an example

- Tasks
  - Specify module 5 days
  - Code module 8 days
  - Test module 6 days
- At the beginning of day 20, PV = 19 days
- If everything but testing, completed EV = 13 days
- Schedule variance (SV) = EV-PV = 13-19 = -6
- Schedule performance indicator (SPI) = EV / PV = 13/19 = 0.68
- SV negative or SPI <1.00, indicates project is behind schedule.





Earned value analysis – actual cost

- Actual cost (AC) is also known as Actual cost of work performed (ACWP)
- In previous example, if
  - 'Specify module' actually took 3 days
  - 'Code module' actually took 4 days
- Actual cost = 7 days
- Cost variance (CV) = EV-AC = 13-7 = 6 days
- Cost performance indicator (CPI) = EV/AC = 13/7 = 1.86
- Positive CV or CPI > 1.00 means project is within budget.





Earned value analysis – actual costs cont ...

- CPI can be used to produce new cost estimate
- Budget at completion (BAC) – current budget allocated to total costs of project
- Estimate at completion (EAC) – updated estimate = BAC/CPI
  - e.g. say budget at completion is £19,000 and CPI is 1.86
  - EAC = BAC/CPI = £10,215
  - Projected costs reduced because work being completed in less time





Earned value analysis - Time Variance Example

- Time variance (TV) – difference between time when specified EV should have been reached and time it actually was
- For example, say an EV of £19000 was supposed to have been reached on 1st April and it was actually reached on 1st July then TV = - 3 months
- TV negative indicates project is running late.





Another Example

- Suppose a project is to be completed in one year at the cost of Rs. 100,000. after 3 months, you realize that the project is 30% complete at a cost of Rs. 40,000. Assess the performance of the project.





Another Example ---- solution

- Planned value (PV) = planned percentage completion of work X Budgeted cost = 25% X Rs.100,000 = Rs. 25,000.
- Earned value (EV) = percentage work actually completed X Budgeted cost = 30% X Rs. 100,000 = Rs. 30,000.
- Cost performance index(CPI) = EV/Actual Cost incurred = EV/AC =Rs.30,000/Rs.40,000 = 0.75
- Schedule performance Index (SPI)= EV/PV =Rs.30,000/Rs.25,000 = 1.2





Another Example ---- solution

- **Assessment of project performance:**

✓ Since CPI <1, the project is over budget. For every rupee spent, we are getting only 0.75 worth of work.

✓ Since SPI >1, it indicates that the project is ahead of schedule.

✓ At this rate, the project will be delivered ahead of schedule, but at over budget.

- **So, corrective action needs to be taken**





Summary

- Discussed monitoring of earned value through
  - Schedule variance (SV)
  - Time variance (TV)
  - Cost variance (CV)
  - Performance ratios
- Explained the above with suitable examples





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.






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